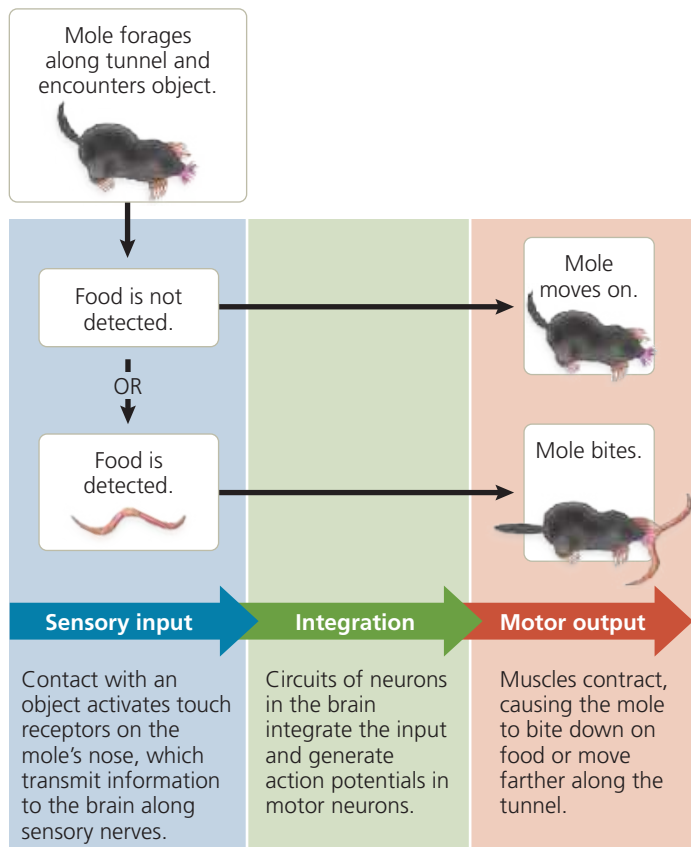


Sensory receptors transduce stimulus energy and transmit signals to the central nervous system

All sensory processes begin with stimuli, and all stimuli represent forms of energy. A sensory receptor converts stimulus energy to a change in membrane potential, thereby regulating the output of action potentials to the central nervous system (CNS). Decoding of this information within the CNS results in sensation.

When a stimulus is received and processed by the nervous system, a motor response may be generated. One of the simplest stimulus-response circuits is a reflex, such as the knee-jerk reflex shown in Figure 49.5. For many other behaviors, sensory input undergoes more elaborate processing. As an example, consider how the star-nosed mole in Figure 50.1 searches for food, or forages, in a tunnel. When the mole's nose contacts an object, touch receptors in the nose are activated (Figure 50.2). These receptors transmit sensory information about the object, such as whether the object is moving, to the mole's brain. Circuits in the brain integrate the input and initiate one of two response pathways. If prey

▼ **Figure 50.2** A simple response pathway: foraging by a star-nosed mole.



or other food is detected, the brain sends motor output commands to skeletal muscles that cause the jaws to bite down. If no food is detected, the brain sends instructions to skeletal muscles to continue movement along the tunnel.

With this overview in mind, let's examine the general organization and activity of animal sensory systems. We'll focus on four basic functions common to sensory pathways: sensory reception, transduction, transmission, and perception.

Sensory Reception and Transduction

A sensory pathway begins with **sensory reception**, the detection of a stimulus by specialized sensory cells. Each sensory cell is either a neuron or a cell that regulates a neuron (Figure 50.3). Some sensory cells exist singly; others are collected in sensory organs, such as the star-shaped nose of the mole in Figure 50.1.

The term **sensory receptor** describes a sensory cell or organ, as well as the subcellular structure that detects stimuli. Some sensory receptors respond to stimuli from within the body, such as blood pressure and body position. Other receptors detect stimuli from outside the body, such as heat, light, pressure, or chemicals. Some of these receptors are sensitive to the smallest possible unit of stimulus. Most light receptors, for example, can detect a single quantum (photon) of light.

Although animals use a range of sensory receptors to detect widely varying stimuli, the effect in all cases is to open or close ion channels. The resulting change in the flow of ions across

▼ **Figure 50.3** Neuronal and non-neuronal sensory receptors.

